Lachlan Watts-Tobin

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EDUCATION

University of California, Berkeley Berkeley, CA
B.S Electrical Engineering & Computer Science, Minor in Mechanical Engineering May 2028
Coursework: Digital Design and Integrated Circuits (With FPGA Lab), Computer Architecture, Internet of Things, Machine Structures, Circuits and Devices, Data Structures, Signals and Information Processing, Solid Mechanics

SKILLS

Languages: Verilog, RISC-V, Chisel, C, C++, Rust, Java, JavaScript, Python, MATLAB, SQL, HTML, CSS
Platforms: Git, KiCad, Altium, LTSpice, ESP32, FPGA, SolidWorks, Fusion 360, OpenCV, PyTorch, Pandas, NumPy

EXPERIENCE

Pano AI San Fransisco, CA
Hardware Engineering Intern May 2026 – August 2026
- Redesigned an edge-computer system layout to fit all electrical components within tight space constraints.
- Authored standard operating procedures and designed a manufacturing plan to scale edge-computer production
- Developed real-time lens-contamination detection in Python/OpenCV using FFT and Laplacian variance analysis.

VISION IV (Remote) Yokohama, Japan
Hardware Engineering Intern May 2025 – July 2025
- Designed and optimized a high-performance component using experimental materials for reliability.
- Analyzed potential failure points to guide data-driven improvements to component performance.

UC Berkeley, College of Engineering Berkeley, CA
Mathematics Instructor June 2025 – August 2025
- Taught pre-calculus to ~100 incoming engineering students in a collaborative learning environment.
- Developed a rigorous pre-calculus curriculum with original lessons and assignments.

PROJECTS

Watts & Cseh Electronics
Co-Founder & Chief Electrical Engineering
- Designed and manufactured analog circuits for guitar effects pedals, inspired by vintage hardwired models.

AI Powered Guitar Pedal
Hackathons at Berkeley, (June 2026)
- Bridged AI to hardware, pushing generated parameters from plain English to an ESP32 DSP device over WiFi.

Automated Trading Card Sorter
UC Berkeley, College of Engineering, (November 2026 – December 2026)
- Engineered a machine that automatically sorts trading cards by their live market value, using CV and scraping.

CS61CPU
UC Berkeley, College of Engineering, (November 2025 – December 2025)
- Designed a functional CPU in Logisim capable of executing 32-bit RISC-V instructions.

Carbon Fiber Filament Winder
Bike Builders of Berkeley, (June 2025 – December 2025)
- Designed and built a fully functional 3-axis carbon fiber filament winder to produce bike tube sections.

Filament Winder G-Code Slicer
UC Berkeley, College of Engineering, (January 2025 – April 2025)
- Developed software to generate G-code from part file parameters and control a filament winder to create it.

EXTRACIRRICULAS

Bike Builders of Berkeley Berkeley, CA
CFRP (Carbon Fiber Reinforced Polymer) Engineer August 2024 – Present
- Continuously develop new techniques for vacuuming and post-processing, which drastically improve quality.
- Presented weekly updates on workflow, testing, and proposed process improvements to the entire team.

Cal Solar Vehicle Berkeley, CA
Electrical Engineer Jan 2026 – Present
- Designed a PCB speed controller for the primary motor and wrote corresponding firmware on the control board.